# Split Complex Lorentzian Dynamics

Part XXVII: Atlas-Generated Topology, Continuous Frame Transitions, and the Ordinary Topological Boundary

From algebraic fibrewise coherence to a uniquely reconstructed local-product topology, with the first genuinely global internal compatibility problem isolated for Part XXVIII

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**Abstract**

Part XXVI reached an unusual boundary. The project already possessed a complete algebraic one-fibre picture: intrinsic positively oriented orthonormal frames, a free and transitive visible action, a free and transitive lifted action, and a two-to-one equivariant projection with kernel $\{\pm 1\}$. It then generalized the ordinary frame geometry to an arbitrary family of oriented Euclidean three-spaces over a base. Fibrewise frame spaces, intrinsic orientation-preserving isometry groups, algebraic local trivializations, and overlap transformations with identity, inverse, and triple-overlap laws were reconstructed. Yet the same task also proved that this algebraic layer was globally trivializable by choice and that the automatically inherited sigma topology was useless for geometric coherence: every fibre became open and arbitrary pointwise trivializations became continuous. The remaining difficulty was therefore no longer hidden in the fibres or in the overlap algebra. It was the topology of variation itself [7, 1].

Task 27 closes that topological deficit without importing a pre-existing vector bundle, frame bundle, principal bundle, or spin structure. It introduces a plain total carrier $E_{\mathrm{tot}} = \sum_{b \in B} E_b$ without an inherited topology and adds a compatible local atlas as genuinely new data. From the atlas it constructs a topology as the supremum of coinduced chart topologies. Equivalently, a subset of the total space is open exactly when it is open in every local product chart. The base projection is continuous, each atlas chart becomes a homeomorphism, and—crucially—the resulting topology is unique among all topologies with those properties. Thus the atlas does not merely coexist with the topology; relative to the given local product data it determines the topology exactly.

The same construction is then carried to the ordinary oriented-frame total space. Fibre trivializations induce frame homeomorphisms, while the fixed model isometry group is equipped only with the minimal topology needed for joint evaluation, inversion, multiplication, and the action on the model frame space. The overlap transformations $g_{ij} : U_i \cap U_j \to G_0^+$ are proved continuous and retain the exact laws

$$g_{ii} = 1, \qquad g_{ji} = g_{ij}^{-1}, \qquad g_{ik} = g_{jk} g_{ij}.$$

The induced frame-coordinate changes are given by the same $g_{ij}$ and are continuous. A concrete constant-family example over $\mathbb{R}$ proves that the atlas-generated topology differs from the automatic sigma topology: individual fibres are open in the latter and not open in the former. The earlier negative diagnosis is therefore promoted to a theorem-level separation of the two notions of topology.

The reverse direction is also closed locally. An arbitrary homeomorphism of frame fibres does not necessarily encode linear metric geometry; Task 27 constructs a non-equivariant frame bijection as a negative control. After equivariance under the model orientation-preserving isometry group is imposed, however, a compatible topological frame trivialization reconstructs a unique orientation-preserving linear isometry of the underlying fibres, including continuity of the reconstructed total-space map. Consequently topological metric-oriented fibre trivializations and equivariant topological frame trivializations are proved equivalent at the local level.

The formal run consists of ten new modules and 2594 Lean lines. The focused Task-27 build is reported green at 8237 jobs, the whole project green at 8308 jobs with zero errors, the new layer has zero warnings, and 67 audited endpoints depend only on the standard baseline `propext`, `Classical.choice`, and `Quot.sound`; there are no proof holes or prohibited proof escapes [2]. Three deliberately open boundaries remain: the converse from one global continuous frame section to a global trivialization is not packaged because joint continuity of the frame transporter is not proved; the atlas-level reverse construction is not packaged independently of a previously generated fibre topology; and no separation properties of the generated total spaces are asserted.

The comparison with established topology is now much sharper. The project has reconstructed the ordinary local-product and transition-function layer expected of an oriented metric rank-three bundle, including continuous structure-group-valued transitions, but it has still not formalized a principal bundle and has not constructed any internal lift over the base [8, 3, 4]. A spin structure remains absent for exactly the right reason: the ordinary continuous transition system is now available, while the separately reconstructed two-sheeted internal group projection has not yet been made coherent with that system [5]. Part XXVIII should therefore begin from the frozen continuous transitions and ask, without assuming a conventional target, when local internal representatives exist, how two such choices differ, and what compatibility condition appears on multiple overlaps. The Conservation of Difficulty has moved again: topology is no longer missing, but global compatibility above continuous ordinary gluing is now exposed as the next mathematical frontier.

**Keywords:** atlas-generated topology; varying metric-oriented fibres; oriented orthonormal frames; local trivialization; continuous transition functions; topological frame family; equivariance; formal verification; spin frontier; Conservation of Difficulty.

# Contents

# 1 The dramatic shift after Part XXVI

The preceding stage did something more important than add a base parameter. It separated two notions that are often silently merged in informal reasoning: a family of isomorphic fibres and a bundle-like object whose fibres vary coherently. At the algebraic level Task 26 had already obtained, for every $b \in B$, a real oriented Euclidean three-space $E_b$, its positively oriented orthonormal frame carrier $\mathcal{F}_b^+$, and the intrinsic orientation-preserving isometry group $G_b^+$ acting freely and transitively on $\mathcal{F}_b^+$. Given local identifications with a fixed model $E_0$, one also obtains overlap elements in the model group $G_0^+$ satisfying the exact algebraic transition laws [1].

That seemed at first close to the familiar bundle picture. The crucial negative result was that it was not yet close enough. Choice produces an algebraic identification $E_b \simeq E_0$ independently at every base point. Hence algebraic trivializations exist even globally. No obstruction can therefore live in the pointwise linear algebra alone. Worse, the automatic topology on the dependent sum makes precisely these arbitrary pointwise choices appear continuous. The intended notion of "nearby fibres vary coherently" had not been formalized at all.

Part XXVII starts at this failure. It does not ask for more algebra upstairs, and it does not introduce a conventional bundle object. Instead it asks:

> What exact local data are sufficient to manufacture the missing topology, and how much of the ordinary frame geometry can then be recovered purely from those data?

The answer closes all three hard targets of Task 27. A compatible atlas generates the total-space topology, determines it uniquely, induces the frame topology and continuous transition system, and supports a reverse reconstruction from equivariant frame charts to metric-oriented fibre charts.

**Methodological principle 1** (Topology is not a label attached to the family)**.** A topology on the base and a fibrewise topological structure do not determine the topology of the total varying object. The missing information is local coherence. In Task 27 this information is supplied by a compatible local-product atlas and only then converted into a global topology.

This is the first stage in the project at which the word "global" becomes genuinely topological rather than merely set-theoretic.

# 2 Formal source, scope, and provenance

The supplied Task-27 archive has SHA-256

`df46fc45474acaa8b015c0ff9f96f4018876076c2ce97b60fe18a8bcbb6f933a`.

The pinned environment is Lean 4.28.0 with Mathlib revision

`8f9d9cff6bd728b17a24e163c9402775d9e6a365`.

The previous Task-26 source was checked against the supplied predecessor archive and is bytewise unchanged. The new source chain is

`ModelTopology` → `AtlasTopology` → `AtlasExtension` → `FiberTopology` → `FrameTopology` → `ContinuousTransitions` → `FrameRightAction` → `ReverseReconstruction` → `TopologicalFrameFamily` → `Task27`.

| Check | Result | Detail |
|---|---|---|
| Focused Task-27 build | PASS | 8237 jobs |
| Whole-project build | PASS | 8308 jobs, 0 errors |

| Check | Result | Detail |
|---|---|---|
| Task-27 warnings | 0 | 114 project-wide warnings are inherited from earlier unchanged layers |
| New source | 10 modules | 2594 Lean lines |
| Audited endpoints | 67 | exposed through `Task27.lean` |
| Axiom baseline | exact | `propext`, `Classical.choice`, `Quot.sound`; some definitions need none |
| `sorry`/`admit` | 0/0 | no proof holes |
| Prohibited escapes | 0 | no custom `axiom`, `implemented_by`, `native_decide`, `bv_decide`, `unsafe`, or `partial` |
| Inherited source | unchanged | Tasks I–XXVI untouched by the new layer |

**Proof boundary 1** (Verification boundary)**.** *The supplied source, audit, endpoint statements, toolchain and manifest, archive hash, prohibited-token scan, module dependencies, and the critical topology and reconstruction proofs were inspected for this paper. The reported Lean build results and `##print axioms` outputs are taken from the supplied formal artifact; the paper-generation environment did not independently rebuild the Lean project.*

The scope firewall remains strict. Task 27 contains no lifted frame family, no lifted transition system, no principal-bundle axiom, no imported vector-bundle or frame-bundle object, no characteristic class, no cohomology, no connection, and no smooth structure. The base is an arbitrary topological space; no Hausdorff, connectedness, paracompactness, second-countability, local-compactness, or manifold assumption is used.

## 3 The total carrier is deliberately topology-free

Let $B$ be a topological space and $E : B \to \mathrm{Type}$ the inherited metric-oriented rank-three family. Task 27 introduces

$$E_{\mathrm{tot}} := \sum_{b:B} E_b$$

through the plain definition `Total`. This design decision is technically small but conceptually central: the declaration is not an abbreviation that allows type-class search to recover the automatic sigma topology. A topology on $E_{\mathrm{tot}}$ must therefore enter explicitly.

The base projection is the set-theoretic map

$$\pi_E : E_{\mathrm{tot}} \longrightarrow B, \qquad (b, v) \longmapsto b.$$

At this stage there is no assertion that $\pi_E$ is continuous, because continuity has no meaning until a topology on $E_{\mathrm{tot}}$ has been chosen.

**Methodological caveat 1** (A base topology is not enough)**.** *Even though every fibre $E_b$ already carries its norm topology and $B$ carries a topology, these data alone do not determine how points in different fibres approach one another. This is exactly the information that was missing in Part XXVI.*

## 4 The atlas as additional mathematical data

The new object is a local-product atlas. Abstractly, a `ChartAtlas` consists of open sets $U_i \subseteq B$ covering $B$, local fibre equivalences

$$\phi_{i,b} : E_b \simeq E_0 \qquad (b \in U_i),$$

and a continuity condition on the induced overlap maps. In the metric-oriented specialization `MetricOrientedFibreAtlas`, the local maps are the algebraic orientation-preserving linear isometries inherited from Task 26.

The atlas is not derived from the bare family. It is new data. This dependency is essential:

$$\boxed{\text{metric-oriented family} + \text{compatible local atlas} \quad \Longrightarrow \quad \text{coherent topology}}$$

and not

$$\text{metric-oriented family} \Longrightarrow \text{local topological triviality.}$$

For a chart indexed by $i$, write

$$\psi_i : U_i \times E_0 \longrightarrow E_{\mathrm{tot}}$$

for the inverse local coordinate map followed by inclusion into the total carrier. Task 27 defines

$$\tau_A := \bigvee_i \operatorname{coinduced}(\psi_i, \tau_{U_i \times E_0}), \tag{1}$$

the supremum of the coinduced chart topologies.

The corresponding theorem `ChartAtlas.isOpen_top_iff` gives the more intuitive characterization:

**Theorem 1** (Atlas-open characterization)**.** *A subset $V \subseteq E_{\mathrm{tot}}$ is open in $\tau_A$ if and only if, for every chart $i$, its representation in the local product $U_i \times E_0$ is open.*

Thus the topology is not guessed externally. It is synthesized from the declared local coherence.

# 5 Existence and uniqueness of the coherence topology

The first hard target is stronger than construction alone. Task 27 proves that the generated topology is admissible: the base projection is continuous and every atlas chart and inverse chart are continuous. Therefore each local coordinate map becomes a genuine homeomorphism

$$\pi_E^{-1}(U_i) \simeq_{\mathrm{top}} U_i \times E_0.$$

More importantly, the topology is unique relative to these atlas data.

**Theorem 2** (Uniqueness relative to an atlas)**.** *Let $\tau$ be any topology on $E_{\mathrm{tot}}$ such that the base projection is continuous and every chart of $A$ is a homeomorphism onto the corresponding local product. Then*

$$\tau = \tau_A.$$

The Lean endpoint is `ChartAtlas.IsAdmissible.eq_top`, with the metric-oriented specialization `MetricOrientedFibreAtlas.eq_top_of_isAdmissible`. The theorem is not definitional: it compares an arbitrary admissible topology with the atlas-generated topology and proves equality.

This creates the first exact reconstruction loop of Part XXVII:

$$\text{atlas} \longrightarrow \text{topology} \longrightarrow \text{charts are homeomorphisms,}$$

while admissibility of the same charts forces the topology back to the unique $\tau_A$.

**Methodological principle 2** (Local data can determine global topology without determining global triviality)**.** The atlas determines how local pieces are glued topologically. It does not imply that a single global product chart exists. The distinction between local coherence and global triviality is therefore preserved rather than erased.

## 6 Genuine topological fibre trivializations

Once a total topology is explicit, the term "continuous trivialization" finally becomes non-vacuous. For an open $U \subseteq B$, `TopologicalFibreTriv` contains the inherited algebraic trivialization together with continuity of the total chart and its inverse. Fibrewise, the maps remain real-linear, isometric, and orientation-preserving.

Task 27 proves equivalence between two formulations:

(i) a pointwise metric-oriented algebraic family together with continuity of its assembled total-space maps;

(ii) a homeomorphism

$$\pi_E^{-1}(U) \simeq_{\text{top}} U \times E_0$$

compatible with the projection and carrying the inherited fibrewise linear-isometric-orientation data.

The forward endpoint is `TopologicalFibreTriv.homeo`; its converse is `topologicalFibreTriv_of_homeo`.

This is the precise Level-2 replacement for Task 26's Level-1 trivializations. At Level 1, choice supplied pointwise identifications on every subset, including all of $B$. At Level 2, continuity is measured in the atlas-generated topology and therefore encodes relations between different base points.

## 7 Why the automatic sigma topology is mathematically wrong

Part XXVI had already shown that the automatic dependent-sum topology makes every fibre open and makes arbitrary algebraic trivializations continuous. Task 27 upgrades that diagnosis into an explicit separation theorem.

Consider a constant family over $B = \mathbb{R}$ with fibre $E_0$ and a single global product chart. In the atlas-generated topology, a fibre

$$\{b_0\} \times E_0$$

is not open. In the automatic sigma topology, the corresponding fibre is open. Therefore

$$\tau_{\text{atlas}} \neq \tau_{\Sigma}. \tag{2}$$

The formal endpoint is `Example27.atlasTop_ne_sigmaTop`.

This result matters beyond one counterexample. It proves that the failure in Part XXVI was not a vague concern about implementation choices. Two different topologies on the same underlying dependent carrier support radically different notions of variation. The intended geometric coherence is encoded by the atlas topology, not by the type-theoretic disjoint-union topology.

**Methodological caveat 2** (No separation theorem is claimed)**.** *The atlas topology is the unique topology compatible with the supplied charts, but Task 27 does not prove that it is Hausdorff, locally compact, or otherwise separated. No such hypothesis was assumed on the base, and none is silently recovered.*

## 8 Minimal topology on the model group

To speak about continuous transition functions, the target model group itself needs a topology. Task 27 deliberately avoids importing a Lie-group development. The model carrier is the fixed Euclidean space

$$E_0 = \mathbb{R}^3,$$

and the orientation-preserving model isometry group is $G_0^+$.

The ambient linear-isometry space $E_0 \simeq_{\text{lin,isom}} E_0$ is equipped with the topology of pointwise convergence, induced from the function space $E_0 \to E_0$. The subgroup $G_0^+$ receives the subtype topology. For a family $g : X \to G_0^+$, continuity is characterized pointwise:

$$g \text{ continuous} \iff \forall m \in E_0, \quad x \mapsto g(x)m \text{ continuous}.$$

From finite-dimensional frame expansions, Task 27 proves joint continuity of evaluation, continuity of inversion, and continuity of multiplication. Thus $G_0^+$ is a topological group at exactly the level needed by the later transition theorems.

The model frame space $\mathcal{F}_0^+$ is similarly given the topology of pointwise convergence of its three frame vectors, and the visible action

$$\mathcal{F}_0^+ \times G_0^+ \longrightarrow \mathcal{F}_0^+$$

is continuous.

**Remark 1.** *The task notes that the chosen topology coincides with more familiar finite-dimensional operator topologies, but no such identification is used. The proof boundary is therefore deliberately narrower than standard Lie-group theory.*

## 9 From fibre topology to frame topology

The ordinary oriented-frame carrier inherited from Task 26 is

$$\mathcal{F}_{\text{tot}}^+ := \sum_{b:B} \mathcal{F}_b^+, \qquad \pi_F : \mathcal{F}_{\text{tot}}^+ \to B.$$

Task 27 constructs its topology from the induced frame atlas rather than inheriting the dependent-sum topology.

A fibre chart $\phi_{i,b} : E_b \to E_0$ sends each oriented orthonormal frame in $E_b$ to an oriented orthonormal frame in $E_0$. Applied to all three frame vectors, this yields a frame chart

$$\pi_F^{-1}(U_i) \simeq_{\text{top}} U_i \times \mathcal{F}_0^+.$$

The task proves continuity of the corresponding overlap maps and then defines `frameTop` by the same atlas-generation mechanism as for the fibre total space.

As before, existence and uniqueness both hold:

- the induced frame atlas is admissible for `frameTop`;
- any other topology making the same charts homeomorphisms and the projection continuous must coincide with `frameTop`.

Moreover, the construction is not restricted to atlas domains. Any topological fibre trivialization over any open $U$ induces a topological frame trivialization over the same $U$:

$$\text{TopologicalFibreTriv}(U) \longrightarrow \text{TopologicalFrameTriv}(U).$$

This is the theorem `topologicalFibreTriv_induces_topologicalFrameTriv`.

## 10 Continuous transition functions

The algebraic overlap element from Task 26 now becomes a continuous map. For two local trivializations $i, j$ and $b \in U_i \cap U_j$, define

$$g_{ij}(b) = \phi_{j,b} \circ \phi_{i,b}^{-1} \in G_0^+.$$

Task 27 proves

$$g_{ij} : U_i \cap U_j \longrightarrow G_0^+$$

is continuous. This continuity is not inferred from the algebraic identities. It is derived from continuity of the local product charts and the topology of $G_0^+$.

The inherited overlap laws survive unchanged:

$$g_{ii}(b) = 1, \tag{3}$$
$$g_{ji}(b) = g_{ij}(b)^{-1}, \tag{4}$$
$$g_{ik}(b) = g_{jk}(b)g_{ij}(b). \tag{5}$$

The order in (5) is part of the formal convention and is preserved consistently throughout the frame-coordinate formulas.

If $v$ is a frame in $E_b$, its two model coordinates satisfy

$$\mathrm{coord}_j(v) = g_{ij}(b) \cdot \mathrm{coord}_i(v). \tag{6}$$

The combined frame transition map on $(U_i \cap U_j) \times \mathcal{F}_0^+$ is continuous.

Thus the ordinary transition system is no longer merely an algebraic record. It has acquired precisely the regularity that Part XXVI was unable to express.

## 11 The reverse direction needs equivariance

The third hard target asks whether topological frame data determine the underlying fibre data. A naive answer would be false: an arbitrary homeomorphism or bijection of the frame carrier can scramble frames in ways unrelated to any linear isometry of the vector space.

Task 27 formalizes this distinction. It defines a right action

$$\mathcal{F}_0^+ \times G_0^+ \longrightarrow \mathcal{F}_0^+$$

and a `CompatibleFrameTriv` as a topological frame trivialization whose fibre maps are equivariant for this action.

The need for equivariance is not merely explanatory. A non-equivariant frame bijection is explicitly constructed as a negative control. Therefore

$$\text{frame-space bijection} \not\Longrightarrow \text{linear metric-oriented fibre map.}$$

With equivariance imposed, the situation changes completely.

**Theorem 3** (Reverse reconstruction)**.** *Let $U \subseteq B$ be open and let a topological frame trivialization over $U$ be equivariant under the model orientation-preserving isometry action. Then it determines a unique fibrewise orientation-preserving linear isometry $E_b \simeq E_0$ for every $b \in U$, and these maps assemble continuously into a topological fibre trivialization.*

The reconstruction uses the fact that a positively oriented orthonormal frame determines its coordinate isometry. Equivariance guarantees that the result does not depend on arbitrary frame choices. Task 27 proves linearity, isometry, orientation preservation, continuity of the total reconstructed map, and uniqueness. The main endpoints are `reconIso`, `reconIso_orientation`, `reconIso_unique`, and `fibreTrivOfCompatibleFrameTriv`.

Consequently the local structures are equivalent:

$$\boxed{\text{Topological metric-oriented fibre trivializations} \simeq \text{equivariant topological frame trivializations}}. \tag{7}$$

The Lean equivalence is `fibreTrivEquivCompatibleFrameTriv`.

**Methodological principle 3** (Representational freedom must be constrained by structure)**.** A frame carrier contains more set-theoretic automorphisms than those arising from the underlying linear geometry. Equivariance is the exact condition used here to remove that excess freedom and recover the fibre map uniquely.

## 12 A compact ordinary topological interface

After the reconstruction results, Task 27 freezes the ordinary topological layer in two compact structures.

`TopologicalFrameFamily` contains the metric-oriented family together with its local atlas. It therefore determines both the fibre and frame total-space topologies, continuous local charts, and the fibrewise free/transitive frame action.

`OrdinaryTransitionSystem` contains the data required for the next stage:

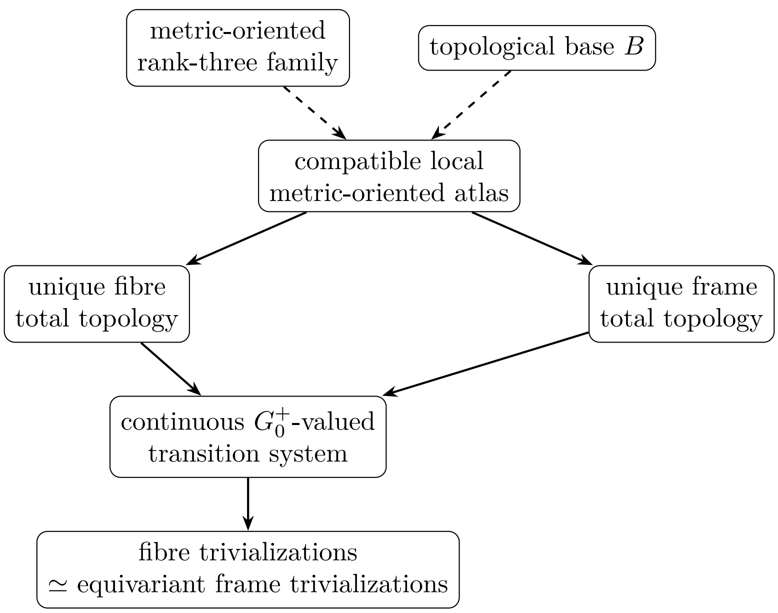
- open sets $U_i$ covering $B$;
- continuous transition maps $g_{ij} : U_i \cap U_j \to G_0^+$;
- identity, inverse, and triple-overlap laws.

The function `TopologicalFrameFamily.toOrdinaryTransitionSystem` produces this frozen output.

This interface is intentionally ordinary. There is no object above the transition maps, no local internal representatives, and no lifted gluing law.

## 13 Exact mathematical dependency map

The new layer separates definitions, extra data, constructions, equivalences, and negative results. The dependency structure can be compressed as follows:



Here the dashed arrows mark genuinely additional data rather than theorem-level consequences. In particular, the atlas is not derived from the bare family. Once the atlas is supplied, the fibre and frame topologies and the continuous transition system are derived.

The most important negative edges are equally explicit:

$$\begin{aligned}
\text{base topology} + \text{fibrewise topology} &\not\Rightarrow \text{coherent total-space topology},\\
\text{pointwise algebraic triviality} &\not\Rightarrow \text{topological triviality},\\
\text{automatic sigma topology} &\not\Rightarrow \text{intended coherence topology},\\
\text{arbitrary frame bijection} &\not\Rightarrow \text{linear metric-oriented fibre map}.
\end{aligned}$$

| Formal endpoint | Mathematical content |
|---|---|
| `MetricOrientedFibreAtlas.top` | atlas-generated topology on the fibre total carrier |
| `MetricOrientedFibreAtlas.eq_top_of_isAdmissible` | uniqueness of every topology making the same atlas admissible |
| `TopologicalFibreTriv.homeo` | genuine local fibre product homeomorphism |
| `Example27.atlasTop_ne_sigmaTop` | formal separation from the automatic sigma topology |
| `MetricOrientedFibreAtlas.frameTop` | induced topology on the ordinary frame total carrier |
| `MetricOrientedFibreAtlas.continuous_transitionFun` | continuity of ordinary transition functions |
| `MetricOrientedFibreAtlas.transitionFun_trans` | exact triple-overlap law |
| `fibreTrivEquivCompatibleFrameTriv` | local equivalence of fibre and equivariant frame trivializations |
| `TopologicalFrameFamily.toOrdinaryTransitionSystem` | frozen continuous ordinary transition interface |

# 14 What has and has not been reconstructed compared with established mathematics

The standard theory of fibre bundles starts from local product charts and compatibility of their overlaps; these charts determine the total-space topology and yield transition functions valued in a structure group [8, 3]. For an oriented metric rank-three vector bundle, restricting to oriented orthonormal frames yields a frame object with orientation-preserving orthogonal transition functions [4].

At this point the present project has reconstructed, from its own dependency chain:

1. a topological base with no manifold assumption;
2. a varying family of real oriented Euclidean three-spaces;
3. an explicit compatible local metric-oriented atlas as additional data;
4. a unique atlas-generated topology on the fibre total carrier;
5. genuine local product homeomorphisms preserving fibrewise linear, metric, and orientation data;
6. a unique induced topology on the oriented-frame total carrier;
7. a topological model group $G_0^+$ and continuous frame action;

8. continuous $G_0^+$-valued transition functions with exact overlap laws;

9. equivalence between local fibre trivializations and equivariant local frame trivializations.

This is already almost all of the ordinary local data that a conventional oriented orthonormal frame-bundle description would organize. Nevertheless the formal project deliberately stops short of asserting that a principal bundle has been constructed. No principal-bundle structure or universal API is imported or proved; the paper therefore describes the result by the exact objects that exist rather than by a stronger categorical label.

The relation to a spin structure is even more constrained. Earlier parts independently reconstructed a two-sheeted internal group projection with kernel $\{\pm 1\}$ and a complete one-fibre lifted torsor [6]. Classical spin geometry requires a global lift of the oriented orthonormal frame structure compatible with the group double cover [5]. Task 27 has now supplied the missing ordinary continuous transition system, but no such lifted system over $B$ is yet present.

Hence the established comparison may be summarized as

| |
|---|
| ordinary local-product topology and continuous frame transitions<br>**reconstructed** |

*but*

| |
|---|
| coherent internal representatives over those transitions<br>**not yet reconstructed** |

# 15 Three exact open boundaries

Task 27 meets its central success criterion, but three boundaries remain and should not be blurred in later interpretation.

## 15.1 Global frame section versus global trivialization

A global topological trivialization implies a continuous global frame section. The converse is not packaged. The missing technical statement is joint continuity of the frame transporter as both source and target frames vary. Pointwise existence and uniqueness of the transporter are already proved.

This is a local completeness issue in the ordinary layer, not the next main obstruction. It does not block the frozen transition-system output.

## 15.2 No independent frame-atlas-first reconstruction

Equation (7) is a complete local equivalence for trivializations relative to the established topologies. The project does not additionally define a frame atlas from scratch, use it to generate a frame topology independently, and then reconstruct the fibre atlas without referring to the fibre-generated topology. Doing so without circularity would require a second foundational route.

The current theorem should therefore be stated precisely as a local equivalence of compatible trivializations, not as an autonomous equivalence between all possible fibre atlases and all possible frame atlases.

## 15.3 No separation properties

No theorem asserts Hausdorffness, local compactness, paracompactness, or analogous global quality of the generated total spaces. Those properties depend on additional assumptions and are outside the current formal scope.

## 16 Conservation of Difficulty

The trajectory from Parts XXV–XXVII makes the Conservation of Difficulty especially visible.

In Part XXV, the hard question was internal and one-fibre: does the lifted frame carrier really carry the expected free/transitive action? The answer was yes. In Part XXVI, the hard question moved outward: how can one even speak of varying fibres without smuggling in a bundle? The answer was to reconstruct the algebraic family first, whereupon the absence of topological coherence became unavoidable. Task 27 now closes that gap by isolating a compatible atlas as the additional datum and deriving the unique topology and continuous ordinary transition system from it.

The difficulty has not vanished. It has moved once more:

$$\begin{array}{ccccc} \text{one-fibre internal algebra} & \longrightarrow & \text{varying algebraic family} & \longrightarrow & \text{ordinary topological gluing} \\ \text{closed} & & \text{closed} & & \text{closed in XXVII} \end{array}$$

while the next unresolved relation is vertical rather than horizontal:

$$\text{continuous ordinary transition data} \quad \overset{?}{\longleftarrow} \quad \text{coherent internal representatives above them.}$$

This shift is mathematically significant. The project no longer lacks the topology required to formulate a global compatibility problem. It now possesses enough topology that such a problem can be asked sharply.

## 17 Outlook: Part XXVIII as a compatibility test above continuous gluing

Part XXVIII should begin only from the frozen ordinary transition system

$$g_{ij} : U_i \cap U_j \longrightarrow G_0^+$$

with continuity and the exact overlap laws already certified. Separately, the project already contains a two-to-one internal group projection

$$p : \mathcal{L} \longrightarrow G_{\mathrm{vis}}$$

with two-element kernel and complete one-fibre local structure. The next task should connect these strands without assuming that the connection succeeds globally.

The natural questions are now precise:

1. Given a continuous ordinary transition map on an overlap, on which open subdomains can one choose a continuous internal representative above it?
2. Is refinement of the ordinary cover sufficient to guarantee such local representatives?
3. If two continuous internal representatives project to the same ordinary transition, what is their exact ratio and what regularity does that ratio inherit?
4. On a triple overlap, how does the product of two chosen internal representatives compare with the representative of the composite ordinary transition?
5. Is the discrepancy forced into the two-element kernel?
6. How does changing local representative choices alter that discrepancy?
7. What exact condition is necessary and sufficient for the internal representatives to obey the same gluing law as the ordinary transitions?

These questions should be posed using only the reconstructed projection, the continuous ordinary transition system, and local continuity. No conventional global object should be introduced as an assumption, and no obstruction invariant should be given to the prover in advance.

**Proof boundary 2** (The XXVIII firewall)**.** *Part XXVII supplies the ordinary topology and continuous overlap system. It does not prove that continuous internal representatives exist on arbitrary whole overlaps, does not construct a global lifted total space, and does not identify any multiple-overlap discrepancy with a preselected classical invariant. Those are new theorem-level questions.*

The expected methodological sequence is therefore

$$\boxed{\begin{array}{c}\text{ordinary continuous transitions}\\ \downarrow\\ \text{local internal representatives}\\ \downarrow\\ \text{comparison on overlaps}\\ \downarrow\\ \text{multiple-overlap compatibility test}\end{array}}$$

with a stop immediately after the strongest exact compatibility criterion has been obtained.

## 18 Engineering provenance and failed-build ledger

The formal audit records ten failed builds during development. All were elaboration or proof-plumbing failures; no mathematical statement was changed.

| # | Module / error class | Diagnosis | Repair |
|---|---|---|---|
| 1 | `AtlasTopology`: unknown indexed topology notation | topology-indexed notation required the proper scope | opened the topology scope |
| 2 | `AtlasTopology`: missing total-space topology instance | a statement-local `letI` did not elaborate later fields as intended | used explicit topology arguments / named instance binders |
| 3 | `FiberTopology`: subtype-product type error | a set used as a type needed the corresponding subtype | replaced by the subtype product |
| 4 | `FiberTopology`: parser error around local scope | scope command ordering was wrong | reordered command and documentation |
| 5 | `FrameTopology`: rewrite motive mismatch | overlap membership had the wrong syntactic form | extracted the required membership fact and beta-reduced before rewriting |
| 6 | `ReverseReconstruction`: missing frame-total topology instance | structure construction could not infer the named binder | installed the atlas frame topology locally |
| 7 | `ReverseReconstruction`: unresolved orientation argument | expected type did not determine the proof argument | supplied inherited orientation preservation explicitly |
| 8 | `ReverseReconstruction`: round-trip reflexivity mismatch | projections of a locally-instantiated term did not reduce at the required transparency | factored a definitional helper and rewrote through it |
| 9 | `Task27`: instance binder used before family binder | binder order prevented synthesis | reordered the binders |
| 10 | `Task27`: compiler IR failure for an alias | a computable alias targeted a noncomputable declaration | removed the alias and exposed the original declaration directly |

The absence of theorem changes matters. Unlike earlier stages in which failed intended statements sometimes exposed real mathematical boundaries, all ten failures here were engineering provenance only. The mathematical frontier described in section 15 is therefore deliberate rather than an artifact of unsuccessful proof attempts.

## 19 Conclusion

Part XXVII closes the topological frontier opened by Part XXVI. A compatible metric-oriented atlas is identified as the genuinely additional datum missing from a bare family of fibres. From that atlas the total-space topology is constructed and uniquely characterized. The frame total space receives its corresponding topology, the model visible group becomes a topological group at the minimal required level, and the ordinary transition transformations become continuous while preserving their exact overlap laws. Finally, equivariance is shown to be the precise condition under which topological frame charts reconstruct the underlying fibre geometry uniquely.

The resulting structure is much closer to the established ordinary frame-bundle picture than any previous stage, but the project still does not possess a spin structure. The reason is now no longer vague. The ordinary topology and continuous gluing system are present; the internal two-sheeted system exists separately; what has not yet been proved is whether and how continuous internal representatives can be chosen and made compatible with the ordinary gluing over the base.

The Conservation of Difficulty has therefore reached a new location. The topology of variation has been reconstructed. The next difficulty is compatibility above that topology.

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